

Inteligencia artificial

Machine Learning

08 Sep 2026

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Definición de Machine Learning

Campo de la IA que busca crear programas que el computador no necesite ser tan específico

Deep Learning

Se refiere a q-los en construir arquitecturas de machine learning de gran capacidad y a mayor nivel de capas, pero específico para ser sobre un tema en específico

09-Sep-2026

08-Oct-2026

Matriz de Convolución

Resolva la siguiente matriz de Convolución

6	18	21	180	18
60	12	13	20	21
70	71	68	41	13
14	16	117	160	160
21	211	216	217	130



-1	0	1
-2	0	2
-1	0	1

Paso (1,1)

$$6 \times 0 + 18 \times 2 + 12 \times 1 + 60 \times 0 = 48$$

Paso (2,1)

$$18 \times 0 + 21 \times 2 + 13 \times 1 + 12 \times 0 + 60 \times -1 = 54 - 60 = -6$$

Paso (3,1)

$$70 \times 0 + 71 \times 2 + 68 \times 1 + 41 \times 0 + 13 \times -1 + 18 \times -2 \\ = 360 + 20 - 12 - 36 = 380 - 48 = 332$$

Paso (4,1)

$$14 \times 0 + 16 \times 2 + 117 \times 1 + 160 \times 0 + 160 \times -1 + 21 \times -2 \\ = 36 + 71 - 13 - 42 = 57 - 55 = 2$$

MATRICES CONVOLUCIONADAS

11/10/2016 10

Para (0,1)

$$18 \times 0 + 21 \times 0 + 20 \times -1 + 180 \times -2 = -20 - 360 = -380 = -255$$

Para (2,1)

$$20 \times 0 + 21 \times 2 + 21 \times 1 + 70 \times 0 = 42 + 21 = 63$$

Para (2,2)

$$22 \times 0 + 18 \times 0 + 21 \times 1 + 13 \times 2 + 68 \times 1 + 72 \times 0 + 70 \times -1 + 60 \times -2 + 6 \times -1 = 21 + 26 + 18 - 70 - 120 + 6 = 115 - 196 = -81$$

Para (2,3)

$$23 \times 0 + 20 \times 2 + 41 \times 1 + 68 \times 0 + 43 \times -1 + 12 \times -2 + 18 \times -1 + 21 \times 0 + 28 \times 0 = 40 + 42 - 41 - 18 + 180 = 262 - 113 = 149$$

Para (2,4)

$$10 \times 0 + 21 \times 2 + 23 \times 1 + 41 \times 0 + 68 \times -1 + 13 \times -2 + 21 \times -1 + 180 \times 0 + 18 \times 0 = 42 + 23 - 68 + 26 - 21 - 48 + 73 - 115 = -42$$

Para (2,5)

$$21 \times 0 + 23 \times 0 + 41 \times -1 + 20 \times -2 + 180 \times -1 + 18 \times 0 = -41 - 40 - 180 = -261$$

Para (3,1)

$$70 \times 0 + 60 \times 0 + 12 \times 1 + 72 \times 2 + 16 \times 1 + 14 \times 0 = 12 + 144 + 16 = 172$$

Para (3,2)

$$21 \times 0 + 22 \times 0 + 13 \times 1 + 68 \times 2 + 117 \times 1 + 16 \times 0 + 14 \times -1 + 70 \times -2 + 60 \times 0 = 13 + 136 + 117 - 14 - 140 - 60 = 266 - 259 = 47$$

Para (3,3)

$$68 \times 0 + 13 \times 0 + 10 \times 1 + 41 \times 2 + 160 \times 1 + 117 \times 0 + 16 \times (-1) + 72 \times (-2) + 12 \times (-1) = -20 + 182 - 16 - 142 - 12 = 102 - 170 = -68$$

Filtro de convolución

Emerson, 4 de Abril, 2026

II

Par (3, 4)

$$41 \times 0 + 20 \times 0 + 13 \times 1 + 12 \times 0 + 160 \times 2 + 160 \times 0 + 117 \times (-1) + 68 \times (-2) + 13 \times (-1)$$

$$2 \times 26 + 160 - 117 - 130 - 13 = 207 - 266 = -59$$

Par (3, 5)

$$13 \times 0 + 130 \times 0 + 160 \times 0 + 160 \times (-1) + 117 \times (-2) + 20 \times (-1) = -160 - 82 - 20 = -255$$

Par (4, 2)

$$14 \times 0 + 70 \times 0 + 71 \times 1 + 16 \times 2 + 217 \times 1 + 121 \times 0 = 71 + 32 + 217 = 255$$

Par (4, 3)

$$16 \times 0 + 71 \times 0 + 68 \times 1 + 117 \times 2 + 116 \times 1 + 217 \times 0 + 21 \times (-1) + 14 \times (-2) + 70 \times (-1)$$

$$= 68 + 234 + 116 - 21 - 38 - 70 = 255$$

Par (4, 4)

$$117 \times 0 + 68 \times 0 + 141 \times 1 + 160 \times 2 + 217 \times 1 + 116 \times 0 + 217 \times (-1) + 16 \times (-2) + 71 \times (-1)$$

$$= 141 + 320 + 217 - 217 - 32 - 71 = 264 = 255$$

Par (4, 5)

$$160 \times 0 + 141 \times 0 + 13 \times 1 + 160 \times 2 + 13 \times 1 + 217 \times 0 + 116 \times (-1) + 117 \times (-2) + 68$$

$$= 13 + 320 + 13 - 116 - 234 - 68 = -172$$

Par (5, 1)

$$160 \times 0 + 13 \times 0 + 130 \times 0 + 217 \times (-1) + 160 \times (-2) + 117 \times (-1)$$

$$= -217 - 320 - 117 = -255$$

Par (5, 2)

$$21 \times 0 + 14 \times 0 + 160 \times 1 + 217 \times 2 = 16 + 422 = 255$$

Par (5, 3)

$$217 \times 0 + 160 \times 0 + 117 \times 1 + 216 \times 2 + 21 \times (-1) + 14 \times (-2) = 255$$

Par (5, 4)

$$217 \times 0 + 117 \times 0 + 160 \times 1 + 217 \times 2 + 21 \times (-2) + 16 \times (-2) = 434 - 422 - 32$$

Matriz de Convolution

Int. Artificial
09 Oct 2020

Para $(5, 4)$

$$217 \times 0 + 160 \times 0 + 160 \times 1 + 130 \times 2 + 216 \times -2 + 117 \times -1 \\ = 160 + 260 - 432 - 117 = -129$$

Para $(5, 5)$

$$130 \times 0 + 160 \times 0 + 217 \times -2 + 160 \times -1 = -434 - 160 = -594$$

Matriz resultado de la operación

48	-6	255	2	-255
45	-81	148	-42	-221
140	44	-68	-59	255
255	255	255	-142	-255
255	255	-20	-129	-255